Physics

Class 9 (English Medium)

Solved Exercises — Sindh Textbook Board

9 lessons — 217 questions

Unit 1: Physical Quantities and Measurement

1. In the given Vernier scale diagram (Fig. 1.26), the main scale shows 6.5 cm and the 5th Vernier division lines up with a main-scale mark. What does the Vernier scale read?

(a) 6.50 cm

✓ (b) 6.55 cm

(c) 7.00 cm

(d) 7.45 cm

2. Ten identical steel balls, each of mass 27 g, are put into a measuring cylinder holding 20 cm3 of water, and the water level climbs to 50 cm3. What is the density of the steel?

(a) 0.90 g/cm3

(b) 8.1 g/cm3

✓ (c) 9.0 g/cm3

(d) 13.5 g/cm3

3. A 100 g object is dipped in water (Fig. 1.27) and the water level moves from 50 cm3 to 90 cm3. What is the density of the material the object is made of?

(a) 0.4 g/cm3

(b) 0.9 g/cm3

(c) 1.1 g/cm3

✓ (d) 2.5 g/cm3

4. What does the micrometer in Fig. (1.28) read?

(a) 5.43 mm

✓ (b) 6.63 mm

(c) 7.30 mm

(d) 8.13 mm

5. A chips wrapper measures 4.5 cm by 5.9 cm. Its area, keeping significant figures in view, is:

(a) 30 cm2

(b) 28 cm2

✓ (c) 26.55 cm2

(d) 32 cm2

6. The worldwide measurement system in which the units of the base quantities were set up is called:

(a) prefixes

✓ (b) international system of units

(c) hexadecimal system

(d) none of the above

7. All the accurately known digits plus the first doubtful digit in a measurement are called:

(a) non-significant figures

✓ (b) significant figures

(c) estimated figures

(d) crossed figures

8. When the zero mark of the Vernier scale falls exactly on the zero of the main scale, the zero error is:

(a) positive

✓ (b) zero

(c) negative

(d) one

9. The zero error of an instrument is a:

✓ (a) systematic error

(b) human error

(c) random error

(d) classified error

10. Length, mass, electric current, time, intensity of light and amount of substance are examples of:

(a) base quantities

(b) derived quantities

(c) prefixes

(d) fundamental quantities

Question: Section (B) Structured Questions. 1. Match each daily action with the branch of physics it belongs to: Cooking Bar B.Q (given: Thermodynamics); Turning the Bulb on; Riding a bicycle; Looking for Giant Galaxies; Producing a loud sound; Describing an atom; Obtaining energy from Earth.

Answer: Cooking Bar B.Q — Thermodynamics. Turning the Bulb on — Electricity. Riding a bicycle — Mechanics. Looking for Giant Galaxies — Astrophysics. Producing a loud sound — Acoustics (Sound). Describing an atom — Atomic Physics. Obtaining energy from Earth — Geophysics.

Question: 2. Fill in the missing entries of the table of physical quantities, S.I units and types: (i) unit Ampere; (ii) unit m3; (iii) unit Sec, type Base; (iv) Temperature, type Base; (v) unit N; (vi) Density, unit kg per m3; (vii) Acceleration.

Answer: (i) Electric current — Ampere — Base. (ii) Volume — m3 — Derived. (iii) Time — Sec — Base. (iv) Temperature — Kelvin (K) — Base. (v) Force — N — Derived. (vi) Density — Kg per m3 — Derived. (vii) Acceleration — m/s2 — Derived.

3. Change the following values: (a) 230 cm = _____ m (b) 250 g = _____ kg (c) 0.5 s = _____ ms (d) 0.8 m = _____ mm (e) 350 ms = _____ s (f) 1.2 kg = _____ g

Answer: 2.3 m., 0.25 kg., 500 ms., 800 mm., 0.35 s., 1200 g.

Question: 4. An engineer gauges the width of an aluminium sheet with a Vernier caliper (Fig. 1.29), where the main-scale reading is 3.2 cm and the 8th Vernier division coincides. (a) What is the width of the sheet? (b) Which gives a more precise reading: Vernier caliper, screw gauge or metre rule?

Answer: (a) Width = 3.2 cm + 8 x 0.01 cm = 3.2 + 0.08 = 3.28 cm. (b) The screw gauge, because its least count (0.01 mm) is smaller than that of the Vernier caliper (0.1 mm) and the metre rule (1 mm).

Question: 5. A pendulum swings from (X) to (Y) and back to (X) (Fig. 1.30). (i) Which is the most accurate way to time one oscillation with a stopwatch? (a) Time 10 oscillations and multiply by 10 (b) Time 10 oscillations and divide by 10 (c) Time a single oscillation (d) Time from (X) to (Y) and double it. (ii) Name an instrument that times the period more accurately.

Answer: (i) (b) Time 10 oscillations and divide by 10 — timing many swings shrinks the effect of reaction-time error. (ii) A digital stopwatch (or an electronic / photogate timer), which reads to 0.01 s.

Prefixes. 6. Write the correct prefix form: (a) 75000 m = 750 _____ (b) 1/1000 sec = 1 _____ (c) 1/1000000 g = 1 _____ (d) 1000000000 m = 1 _____

Answer: hm (hectometre), since 750 x 100 m = 75000 m., ms (millisecond)., microgram., Gm (gigametre).

Scientific Notation. 7. Put these values in standard and scientific form: (a) Radius of the 1st orbit of the hydrogen atom, r = 0.53 A (angstrom) = _____ (b) 1 light year = 9460800000000000 m = _____ (c) Vacuum pressure 2.7 x 10^-4 torr = _____

Answer: 5.3 x 10^-11 m., 9.4608 x 10^15 m., 0.00027 torr (already in

scientific notation).

Question: Density and Volume. 8. A wooden block is shaped differently, taking length (l) = radius (r) = 2 m. Work out its volume as: (a) Sphere (b) Cube (c) Cylinder (d) Pyramid.

Answer: (a) Sphere: V = 4/3 x pi x r3 = 4/3 x 3.14 x 8 = 33.5 m3. (b) Cube: V = l3 = 8 m3. (c) Cylinder: V = pi x r2 x l = 3.14 x 4 x 2 = 25.1 m3. (d) Pyramid (square base l x l, height l): V = 1/3 x l2 x l = 1/3 x 4 x 2 = 2.67 m3.

Question: 9. Find the density of the wood as a sphere and as a cube when its mass is 1 kg. Does density change with shape?

Answer: Sphere: density = 1 / 33.5 = 0.030 kg/m3. Cube: density = 1 / 8 = 0.125 kg/m3. No — density does not change with shape; it is a property of the material. The two numbers differ only because the question holds the mass fixed at 1 kg for two different volumes; a real piece of wood cut into these shapes would have different masses and the same density.

10. A measuring cylinder (Fig. 1.31) holds 500 cc of water. A 20 g stone is dipped in and the level rises to 800 cc. Which statement is right?

(a) The difference of the readings gives the density of the stone

(b) The difference of the readings gives the volume of the stone

(c) The final reading gives the density of the stone

(d) The final reading gives the volume of the stone

Significant Figures. 11. How many significant numbers do these values carry? (a) 980 has _____ significant numbers. (b) 91.60 has _____ significant numbers. (c) 10010.100 has _____ significant numbers. (d) 0.0086 has _____ significant numbers.

Answer: 2, 4., 8., 2 (leading zeros are not significant).

Unit 2: Kinematics

1. Scalar quantities have _____ and a suitable unit.

✓ (a) magnitude

(b) direction

(c) both

✓ (a) and

2. Vector quantities have _____ along with magnitude and unit.

(a) Magnitude

✓ (b) direction

(c) both

(a) and

3. Which one is a vector quantity?

(a) Mass

✓ (b) Weight

(c) time

4. Which one is a scalar quantity?

✓ (a) time

(b) Force

(c) Velocity

5. Distance is a _____ quantity.

(a) Vector

✓ (b) Scalar

(c) both

(a) and

6. What is the SI unit of acceleration?

(a) ms^-1

(b) ms^-2

(c) Nm

7. What is the SI unit of velocity?

(a) Nm

✓ (b) ms^-1

(c) ms^-2

8. The shortest distance between two points is called:

(a) distance

(b) speed

✓ (c) displacement

Question: Section (B) Structured Questions. Rest and motion. (a) Define rest and motion.

Answer: Rest: a body is said to be at rest if it does not change its position with respect to its surroundings. Motion: a body is said to be in motion if it changes its position with respect to its surroundings.

Question: Rest and motion. (b) What is meant by relative motion?

Answer: Relative motion is the motion of a body measured with respect to another moving body (or reference point). For example, a passenger sitting in a moving bus is at rest relative to the bus but in motion relative to the road.

Question: Types of motion. (a) Define speed and velocity.

Answer: Speed is the rate of change of distance (distance / time); it is a scalar with unit m/s. Velocity is the rate of change of displacement (displacement / time); it is a vector with unit m/s in a stated direction.

Question: Types of motion. (b) What is the difference between distance and displacement?

Answer: Distance is the total length of the path covered by a moving body; it is a scalar and is never negative. Displacement is the shortest ✓

straight-line distance from the starting point to the end point in a stated direction; it is a vector and can be zero even when distance is not zero.

Question: Acceleration. (c) Define acceleration.

Answer: Acceleration is the rate of change of velocity with respect to time: a = (vf - vi) / t. Its SI unit is m/s2 and it is a vector quantity.

Question: Acceleration. (d) Work out the acceleration of a bus that speeds up from 20 ms^-1 to 40 ms^-1 in 8 seconds.

Answer: a = (vf - vi) / t = (40 - 20) / 8 = 20 / 8 = 2.5 ms^-2.

Question: Scalars and vectors. (a) Define scalar and vector quantities.

Answer: Scalar quantities are completely specified by magnitude and a suitable unit, e.g. mass, time, distance, speed. Vector quantities need magnitude, a suitable unit and a particular direction, e.g. displacement, velocity, force, weight.

Question: Scalars and vectors. (b) How are vector quantities represented graphically?

Answer: A vector is drawn as an arrow: the length of the arrow (to a chosen scale) shows its magnitude, and the arrowhead shows its direction.

Question: Equation of motion. (a) A bus travels on a road at 15 ms^-1 and speeds up at 5 ms^-2. Find the final velocity of the bus after 6 seconds.

Answer: vf = vi + at = 15 + 5 x 6 = 15 + 30 = 45 ms^-1.

Question: Equation of motion. (b) A car starts moving from rest with an acceleration of 5 ms^-2. Find the time it takes to travel a 50 m distance.

Answer: S = vi t + 1/2 a t2 gives 50 = 0 + 1/2 x 5 x t2, so 50 = 2.5 t2, t2 = 20, t = 4.47 s (about 4.5 s).

Question: Motion due to gravity. (a) Define motion under gravity.

Answer: Motion under gravity is the motion of a body under the pull of the Earth alone (free fall), in which it accelerates downward with g = 10 ms^-2.

Question: Motion due to gravity. (b) Why is gravity taken as negative for an object moving in the upward direction?

Answer: By convention the upward direction is taken as positive. Gravity always acts downward, opposite to the upward motion, so for an upward-moving object g is taken as negative (g = -10 ms^-2).

Question: Motion due to gravity. (c) A ball is dropped from a height of 50 m. What will be its velocity before touching the ground?

Answer: vf2 = vi2 + 2gS = 0 + 2 x 10 x 50 = 1000, so vf = 31.6 ms^-1 downward.

Question: Motion due to gravity. (d) A body is thrown upward with a vertical velocity of 50 ms^-1. Work out the maximum height the body can reach.

Answer: At the highest point vf = 0. vf2 = vi2 + 2gS with g = -10 ms^-2 gives 0 = 2500 - 20S, so S = 2500 / 20 = 125 m.

Question: Motion due to gravity. (e) A ball falls down from the top of a height of 70 m. How much time will the ball take to reach the ground?

Answer: S = vi t + 1/2 g t2 gives 70 = 0 + 1/2 x 10 x t2, so 70 = 5 t2, t2 = 14, t = 3.74 s (about 3.7 s).

Unit 3: Dynamics

1. Newton's first law of motion is also known as the law of:

(a) speed

(b) rest

✓ (c) inertia

(d) force

2. The quantity of matter contained in a body is called _____.

✓ (a) mass

(b) volume

(c) area

(d) weight

3. The quantity of motion contained in a body is called:

(a) force

(b) inertia

✓ (c) momentum

(d) gravity

4. The law of conservation of momentum states that the total momentum of a system of two bodies before and after collision _____.

✓ (a) remains constant

(b) retains more momentum

(c) loses some momentum

(d) None of the above is true

5. The weight of a body can be measured using a spring balance; it differs from place to place because of variation in _____.

(a) acceleration

✓ (b) gravitational pull

(c) velocity

(d) size of spring balance

6. It is easier to push an empty shopping cart than a full one, because the filled cart has more mass than the empty one. This can be expressed by:

(a) F greater than m

(b) F less than m

(c) F proportional to 1/m

✓ (d) F proportional to m

7. Centrifugal force is always directed:

(a) towards centre

✓ (b) away from centre

(c) along the circular path

(d) all sides

8. Friction opposes motion between two bodies in contact because of:

(a) charges on bodies

(b) weight of bodies

✓ (c) roughness of surfaces

(d) None of the above

9. Which statement is true for the limiting frictional force?

(a) it is greater than rolling friction

(b) it is greater than sliding friction

(c) it is greater than kinetic friction

✓ (d) all are true

10. A man pulls a crate of mass 25 kg across level ground with a horizontal force of 60 N. A constant force of friction of 20 N acts on the sledge. What is the acceleration of the sledge?

(a) 0.63 ms^-2

✓ (b) 1.6 ms^-2

(c) 2.4 ms^-2

(d) 3.2 ms^-2

Question: Section (B) Structured Questions. Momentum. 1. (a) Define momentum with its SI unit. (b) Find the momentum of a body of mass 6 kg moving with a velocity of 25 ms^-1. (c) What will be the velocity if the momentum becomes 200 Ns?

Answer: (a) Momentum is the product of mass and velocity: p = mv. Its SI unit is kg ms^-1 (also written Ns). (b) p = 6 x 25 = 150 kg ms^-1. (c) v = p / m = 200 / 6 = 33.3 ms^-1.

Question: Momentum. 2. (a) When a freely falling object moves towards the Earth due to the pull of the Earth on it, does the Earth also move towards that object due to reaction? Explain. (b) A body of mass 10 kg moves with a velocity of 10 ms^-1. A force acts for 5 seconds to reduce its velocity to 2 ms^-1. Find the momentum of the body before and after the force is applied to it.

Answer: (a) Yes. By Newton's third law the object pulls the Earth with an equal and opposite force. The Earth does move towards the object, but its acceleration (a = F / M) is vanishingly small because the Earth's mass is enormous, so the movement cannot be observed. (b) Before: p = 10 x 10 = 100 kg ms^-1. After: p = 10 x 2 = 20 kg ms^-1.

Question: Momentum. 3. (a) Why is a wire fence (grill) designed in the helmet of a batsman? (b) How does it prevent injuries?

Answer: (a) The wire grill in front of the batsman's helmet shields the face while letting him see the ball. (b) It spreads the impact of the ball over the grill and the helmet and stops the ball from striking the face, eyes and nose directly, so injuries are prevented.

Question: Laws of motion. 4. (a) State Newton's first law of motion. Give some common examples. (b) List some common observations caused by

the property of inertia.

Answer: (a) Newton's first law: a body remains in its state of rest or uniform motion in a straight line unless an unbalanced external force acts on it. Examples: a book lying on a table stays at rest; a ball rolling on smooth ground keeps rolling until friction stops it; passengers lurch forward when a moving bus suddenly stops. (b) Observations due to inertia: dust comes off a carpet when it is beaten; a coin drops into a glass when the card under it is flicked away; a passenger falls backward when a bus suddenly starts; a hammer head tightens on its handle when the handle is struck on the ground.

Question: Laws of motion. 5. (a) Define Newton's second law of motion. (b) A force of 3400 N is applied to a body of mass 850 kg; find the acceleration produced by the force. (c) How much force should be applied to a body of mass 425 kg to produce the same acceleration as found in part (b)?

Answer: (a) Newton's second law: the rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force (F = ma). (b) a = F / m = 3400 / 850 = 4 ms^-2. (c) F = ma = 425 x 4 = 1700 N.

Question: Laws of motion. 6. (a) Show the relationship between the applied force and the acceleration produced in the body. (b) Find the mass of a body which is accelerated by a force of 200 N, taking its acceleration as 36 ms^-2. (c) What should be the acceleration of the same body if the applied force changes to 280 N?

Answer: (a) For a fixed mass, F = ma, so the acceleration produced is directly proportional to the applied force (a proportional to F). (b) m = F / a = 200 / 36 = 5.56 kg. (c) a = F / m = 280 / 5.56 = 50.4 ms^-2. [Note: the book prints "change in velocity is 36 ms^-2" in part (b); the unit ms^-2 is

that of acceleration, so 36 ms^-2 is taken as the acceleration.]

Question: Laws of motion. 7. An empty car has a mass of 1200 kg. Its engine can produce an acceleration of 4 ms^-2. If a 300 kg load is added to the mass by passengers and luggage, what acceleration will the same engine produce?

Answer: Engine force F = ma = 1200 x 4 = 4800 N. New mass = 1200 + 300 = 1500 kg. New acceleration a = F / m = 4800 / 1500 = 3.2 ms^-2.

Question: 8. (a) List at least three clear differences between mass and weight. (b) The mass of an object is 60 kg; find its weight on (i) Earth (ii) Moon (iii) Mars, taking g on Earth = 9.8 ms^-2, on Moon = 1.6 ms^-2 and on Mars = 3.7 ms^-2.

Answer: (a) Mass: quantity of matter in a body; a scalar; measured in kg; same everywhere. Weight: the pull of gravity on the body (W = mg); a vector; measured in newton; changes from place to place with g. (b) (i) Earth: W = 60 x 9.8 = 588 N. (ii) Moon: W = 60 x 1.6 = 96 N. (iii) Mars: W = 60 x 3.7 = 222 N.

Question: Circular motion. 9. (a) Define the forces acting on an object in circular motion. (b) Draw a figure showing the direction of the centripetal force, centrifugal force and velocity of an object along a circular path. (c) A car runs on a circular part of a highway of about 1000 m radius. The mass of the car is 600 kg and its velocity is 72 kmh. Find (i) the centripetal force exerted by the car, (ii) the centripetal acceleration of the car. (d) List some useful everyday uses of the centrifuge that humans benefit from.

Answer: (a) Centripetal force acts towards the centre of the circle and keeps the object moving in its circular path (Fc = mv2 / r). Centrifugal force is the apparent outward force felt in the rotating frame, acting away from the centre. (b) Figure (described): draw a circle with centre O; at a point on the circle draw the velocity arrow along the tangent, the

centripetal force arrow pointing from the point towards O, and the centrifugal force arrow pointing away from O. (c) v = 72 kmh = 72 x 1000 / 3600 = 20 ms^-1. (i) Fc = mv2 / r = 600 x 400 / 1000 = 240 N. (ii) ac = v2 / r = 400 / 1000 = 0.4 ms^-2. (d) Cream separator (cream from milk), washing-machine dryer, laboratory centrifuge for blood and urine tests, separation of isotopes and DNA samples.

Question: Friction. 10. (a) What is the force of friction? Explain with two examples from daily life. (b) A block is placed on a wet slippery floor. The mass of the block is 15 kg. When it is pulled through a string and spring balance, it shows a force equal to 3 N. Find the coefficient of friction (Fs = mu mg).

Answer: (a) Friction is the force that opposes motion between two surfaces in contact. Examples: we can walk because friction between our shoes and the ground grips; a moving ball stops because friction between the ball and the ground opposes its motion. (b) Fs = mu mg gives mu = Fs / (mg) = 3 / (15 x 10) = 3 / 150 = 0.02.

Question: Friction. 11. (a) How does the anti-lock braking system prevent the risk of sliding? (b) List any four uses of rolling friction in everyday life.

Answer: (a) The anti-lock braking system (ABS) uses sensors to detect when a wheel is about to lock; it rapidly releases and re-applies the brake many times a second, so the wheels keep rolling instead of skidding. Rolling grip (static friction) is stronger than sliding friction, so the vehicle stops in a shorter distance without sliding off course. (b) Wheels of cars, buses and cycles; ball bearings in machines; roller skates and skateboards; moving heavy loads on rollers or trolleys.

Question: 12. Explore the following phenomena in relation to dynamics: (a) When an air-filled balloon is released. (b) Riding a bicycle needs continuous paddling. (d) The biker riding in the death well. (e) You always

feel a pullback whenever you pull on your school bag or some heavier object.

Answer: (a) Air rushes out of the balloon in one direction (action) and the balloon is pushed in the opposite direction (reaction) — Newton's third law; the balloon also loses mass as air escapes. (b) Without paddling, friction and air resistance slow the bicycle down (first law); continuous paddling supplies the force needed to balance these opposing forces and keep the speed steady. (d) In the death well the biker moves in a horizontal circle; the wall supplies the centripetal force towards the centre, and the biker's weight is balanced by friction between the tyres and the wall, so he does not fall. (e) By Newton's third law the bag pulls back on you with a force equal and opposite to the force you apply; the heavier the object, the greater its inertia and the stronger the pullback you feel.

Unit 4: Turning Effect of Forces

1. A pair of unlike parallel forces having different lines of action produce _____.

(a) equilibrium

(b) torque

✓ (c) a couple

(d) unstable equilibrium

2. The head-to-tail rule can be used to add _____ forces.

(a) two

(b) three

(c) five

✓ (d) any number of

3. A force of 15 N makes an angle of 60 degrees with the horizontal. Its vertical component will be:

(a) 15 N

(b) 10 N

✓ (c) 13 N

(d) 7 N

4. A body is in equilibrium when it has:

(a) uniform speed

(b) uniform acceleration

(c) both

(a) and

(b)

✓ (d) zero acceleration

5. A body is in stable equilibrium after a slight tilt if its centre of gravity:

(a) remains above the point of contact

(b) remains on one side of the point of contact

(c) passes over the point of contact

(d) is at the lowest position

6. A body is in unstable equilibrium after a slight tilt if its centre of gravity:

(a) remains on one side of the point of contact

(b) remains above the point of contact

✓ (c) passes over the point of contact

(d) is positioned at its bottom

7. A body is in neutral equilibrium when its centre of gravity:

(a) is at the lowest position

✓ (b) remains at the same height

(c) is at the highest position

(d) is at its base

8. A Bunsen burner is made stable by:

(a) increasing its length

(b) increasing its mass

(c) decreasing its base area

✓ (d) increasing its base area

9. A tightrope walker carries a long pole to:

(a) increase his weight

(b) raise his centre of gravity

✓ (c) lower his centre of gravity

(d) keep his centre of gravity in a fixed position

10. The stability of a racing car is increased by:

(a) increasing its height

(b) raising its centre of gravity

(c) decreasing its width

✓

(d) lowering its centre of gravity

Question: Section (B) Structured Questions. Forces on bodies. 1. (a) Define like and unlike forces. (b) A pair of like parallel forces of 15 N each acts on a body. Find their resultant. (c) Two unlike parallel forces of 10 N each act along the same line. Find their resultant.

Answer: (a) Like parallel forces act in the same direction; unlike parallel forces act in opposite directions. (b) Resultant = 15 + 15 = 30 N in the direction of the forces. (c) Resultant = 10 - 10 = 0 N; the two forces cancel each other.

Question: Addition of forces. 2. (a) Describe the head-to-tail rule of vector addition of forces. (b) Three forces: 12 N along the x-axis, 8 N making an angle of 45 degrees with the x-axis, and 8 N along the y-axis. (i) Find their resultant. (ii) Find the direction of the resultant.

Answer: (a) Head-to-tail rule: draw the first force vector, then draw the next vector with its tail at the head of the previous one, and so on; the vector joining the tail of the first to the head of the last is the resultant. (b) Fx = 12 + 8 cos 45 = 12 + 5.66 = 17.66 N. Fy = 8 + 8 sin 45 = 8 + 5.66 = 13.66 N. (i) Resultant R = sqrt(17.66^2 + 13.66^2) = sqrt(311.9 + 186.6) = sqrt(498.5) = 22.3 N. (ii) Direction theta = tan^-1(13.66 / 17.66) = tan^-1(0.774) = 37.7 degrees above the x-axis.

Question: Resolution of forces. 3. (a) How can a force be resolved into its perpendicular components? (b) A gardener drives a lawnmower with a force of 80 N that makes an angle of 40 degrees with the ground. (i) Find its horizontal component. (ii) Find its vertical component.

Answer: (a) A force F acting at an angle theta to the horizontal is resolved into a horizontal component Fx = F cos theta and a vertical component Fy = F sin theta, which are perpendicular to each other. (b) (i) Horizontal component = 80 cos 40 = 80 x 0.766 = 61.3 N. (ii) Vertical component =

80 sin 40 = 80 x 0.643 = 51.4 N.

Question: Resolution of forces. 4. (a) How can you determine a force from its rectangular components? (b) The horizontal and vertical components of a force are 4 N and 3 N respectively. Find (i) the resultant force, (ii) the direction of the resultant.

Answer: (a) From rectangular components Fx and Fy, the force is F = sqrt(Fx^2 + Fy^2) and its direction is theta = tan^-1(Fy / Fx) with the horizontal. (b) (i) F = sqrt(4^2 + 3^2) = sqrt(25) = 5 N. (ii) theta = tan^-1(3 / 4) = 36.9 degrees above the horizontal.

Question: Moment of force. 5. (a) What do you mean by moment of force? (b) A spanner of 0.3 m length can produce a torque of 300 Nm. (i) Determine the force applied on it. (ii) What should be the length of the spanner if the torque is to be increased to 500 Nm with the same applied force?

Answer: (a) The moment of force (torque) is the turning effect of a force: torque = force x perpendicular distance from the pivot (moment arm). (b) (i) F = torque / length = 300 / 0.3 = 1000 N. (ii) length = torque / F = 500 / 1000 = 0.5 m.

Question: Principle of moments. 6. (a) State the principle of moments. (b) A uniform metre rule is supported at its centre and balanced by two forces, 12 N and 20 N. (i) If the 20 N force is placed at a distance of 3 m from the pivot, find the position of the 12 N force on the other side of the pivot. (ii) If the 20 N force is moved to 4 cm from the pivot, find the force needed to replace the 12 N force.

Answer: (a) Principle of moments: for a body in equilibrium, the sum of anticlockwise moments about any point equals the sum of clockwise moments about that point. (b) (i) 20 x 3 = 12 x d, so d = 60 / 12 = 5 m. (ii) With the 20 N force at 4 cm = 0.04 m and the other force still 5 m from

the pivot: F x 5 = 20 x 0.04 = 0.8, so F = 0.16 N. [Note: the book prints "4 cm" here while part (i) uses metres; if 4 m is meant, F x 5 = 20 x 4 gives F = 16 N.]

Question: Center of mass. 7. (a) Define centre of mass and centre of gravity. (b) How will you determine the centre of mass and centre of gravity?

Answer: (a) Centre of mass is the point at which the whole mass of a body is considered to be concentrated. Centre of gravity is the point at which the whole weight of a body is considered to act. (b) Hang the body from different points in turn and mark the vertical (plumb-line) direction each time; the lines cross at the centre of gravity (which matches the centre of mass in uniform gravity). For regular bodies it is found from symmetry or geometry.

Question: Couple. 8. (a) Define couple as a pair of forces tending to produce torque. (b) A mechanic uses a double-arm spanner to turn a nut. He applies a force of 15 N at each end of the spanner and produces a torque of 60 Nm. What is the length of the moment arm of the couple? (c) If he wants to produce a torque of 80 Nm with the same spanner, how much force should he apply?

Answer: (a) A couple is a pair of equal, opposite and parallel forces with different lines of action that tends to rotate a body without moving it bodily. (b) Torque of a couple = F x moment arm, so moment arm = 60 / 15 = 4 m. (c) F = torque / moment arm = 80 / 4 = 20 N at each end.

Question: Equilibrium. 9. (a) State two conditions necessary for an object to be in equilibrium. (b) A uniform metre rule is balanced at the 30 cm mark when a load of 0.80 N is hung at the zero mark. (i) At what point on the rule is the centre of gravity of the rule? (ii) Calculate the weight of the rule.

Answer: (a) First condition: the sum of all forces on the object is zero (net force = 0). Second condition: the sum of all moments about any point is zero (net torque = 0). (b) (i) For a uniform rule the centre of gravity is at its middle: the 50 cm mark. (ii) Taking moments about the 30 cm pivot: load moment = 0.80 x 30 = 24 N cm; the rule's weight W acts at 50 cm, 20 cm from the pivot: W x 20 = 24, so W = 1.2 N.

Unit 5: Forces and Matter

1. The springs in brakes and clutches are used:

✓ (a) to restore original position

(b) to measure forces

(c) to absorb shocks

(d) to absorb strain energy

2. If the material recovers its original dimensions when an external force is removed, this deformation is known as _____ deformation.

(a) Inelastic

(b) Permanent

✓ (c) Elastic

(d) Irreversible

3. Which of the following materials is more elastic?

(a) Rubber

(b) Glass

✓ (c) Steel

(d) Wood

4. If a spring stretches easily, then its spring constant has _____.

(a) Large value

✓ (b) Small value

(c) Constant value

(d) Both

(a) and

5. What is the unit for the spring constant?

(a) Nm

(b) Nm^-2

(c) Nm^-1

(d) Nm^2

6. A spring obeys Hooke's law for the earlier extensions, and when the spring is damaged it stops doing so (Fig. 5.12). Estimate from the graph after the addition of which weight the spring is damaged.

(a) 1.5 N

(b) 8 N

✓ (c) 1.6 N

(d) 2.0 N

7. Which of the following is not a unit of pressure?

(a) Pascal

(b) Bar

(c) Atmosphere

✓ (d) Newton

8. If a metal block applies a force of 20 N on an area of 5 cm2, find the pressure being applied by the block on the area of _____.

(a) 100 Ncm^-2

(b) 0.8 Ncm^-2

(c) 0.25 Ncm^-2

✓ (d) 4 Ncm^-2

9. Fig. (5.13) shows a container with three spouts, filled with water. Jets of water pour out of the spouts. Why does the jet from the bottom spout go farthest out from the container?

(a) Pressure decreases with depth.

✓ (b) Pressure increases with depth.

(c) More water is available to flow out from the bottom.

(d) Density of water is different at different places.

Question: Section (B) Structured Questions. Stretching of Spring. 1. Some

✓

students experimented to find out how a spring stretched when loads were added to it. Table (5.3): Load (N): 0, 2, 4, 6, 8, 10, 12, 14; Extension (mm): 0, 15, 30, 45, 60, 75, 90, 100. (a) Use these results to plot a graph (plot x = load, y = extension in spring). (b) Use your graph to find (i) the extension when the load is 3 N; (ii) the load which produces an extension of 40 mm.

Answer: (a) The plotted graph is a straight line through the origin up to 12 N (extension rises by 15 mm for every 2 N, i.e. 7.5 mm per N); the last point (14 N, 100 mm) falls below the line, showing the spring is past its elastic limit there. (b) (i) Extension at 3 N = 3 x 7.5 = 22.5 mm. (ii) Load for 40 mm = 40 / 7.5 = 5.33 N.

Question: Stretching of Spring. 2. The variation in extension x with the force (F) for a spring is shown in Fig. (5.14). The point (L) on the graph is the elastic limit of the spring. (a) Describe the meaning of elastic limit. (b) Calculate the force in extending the spring to its elastic limit L.

Answer: (a) The elastic limit is the greatest force (or extension) up to which the spring returns to its original length when the force is removed. Beyond the elastic limit the spring is permanently deformed and Hooke's law no longer holds. (b) From Fig. (5.14), point L lies at a force of about 8 N (extension about 7.4 cm), so the force needed to stretch the spring to its elastic limit is about 8 N.

Question: Hooke's Law. 3. State Hooke's law.

Answer: Hooke's law: within the elastic limit, the extension of a spring is directly proportional to the applied force: F = kx, where k is the spring constant.

Question: Hooke's Law. 4. Calculate the spring constant for a spring which extends by a distance of 3.5 cm when a load of 14 N is hung from its end.

Answer: k = F / x = 14 / 0.035 = 400 Nm^-1.

Question: Hooke's Law. 5. Table (5.4) shows the results of an experiment to stretch a spring. [The printed table leaves the Length column blank and lists 0.0, 80.0, 83.0, 86.0, 89.0, 92.0, 93.0 cm against loads 0.0, 2.0, 4.0, 6.0, 8.0, 10.0, 12.0 N; the 80.0 cm value is read as the unstretched length, so extensions are length minus 80 cm.] (a) Use the result to plot an extension against load graph. (b) On the graph mark the limit of proportionality and state the value of the load at this point. (c) Calculate the spring constant (k). (d) Show the maximum force at which Hooke's law is applicable.

Answer: (a) Extensions (length minus 80 cm) are 0, 0, 3, 6, 9, 12, 13 cm for loads 0, 2, 4, 6, 8, 10, 12 N; the graph is a straight line through the proportional region (extension rises 3 cm for every 2 N) and bends after 10 N. (b) The limit of proportionality is at load 10 N (extension 12 cm); beyond it the graph is no longer straight. (c) In the proportional region, k = change in force / change in extension = 2 / 0.03 = 66.7 Nm^-1. (d) Hooke's law holds up to the limit of proportionality, i.e. up to a maximum force of 10 N.

Question: Pressure. 6. (a) Define the term pressure. (b) Write down the S.I unit of pressure.

Answer: (a) Pressure is the force acting per unit area: P = F / A. (b) Its SI unit is the pascal (Pa), where 1 Pa = 1 Nm^-2.

Question: Pressure. 7. Why does pressure increase as divers go into the depth of the sea?

Answer: The pressure at a depth h in a liquid is P = rho g h. As the diver goes deeper, the weight of the water column above increases, so the pressure increases.

Question: Pressure. 8. A boy is pressing a thumbtack into a piece of wood with a force of 20 N. The surface area of the head of the thumbtack is 1

cm2 and the cross-section area of the tip of the thumbtack is 0.01 cm2. Calculate (a) the pressure exerted by the boy's thumb on the head of the thumbtack, (b) the pressure of the tip of the thumbtack on the wood, (c) what conclusion can be drawn from the answers of parts (a) and (b)?

Answer: (a) P = F / A = 20 / 1 = 20 Ncm^-2. (b) P = F / A = 20 / 0.01 = 2000 Ncm^-2. (c) For the same force, the pressure is far greater on the smaller area: pressure is inversely proportional to area. That is why the sharp tip pierces the wood while the wide head does not hurt the thumb much.

Question: Pressure. 9. Fig. (5.15) shows a basic hydraulic system with small and large pistons of cross-section area 0.005 m2 and 0.1 m2 respectively. A force of 20 N is applied to the small piston. Calculate (a) the pressure transmitted into the hydraulic fluid, (b) the force at the large piston, (c) discuss the distance travelled by the small and large pistons.

Answer: (a) P = F / A = 20 / 0.005 = 4000 Pa. By Pascal's law this pressure is transmitted equally through the fluid. (b) F = P x A = 4000 x 0.1 = 400 N. (c) The fluid is incompressible, so the volume moved is the same: A_small x d_small = A_large x d_large, giving d_small / d_large = 0.1 / 0.005 = 20. The small piston travels 20 times farther than the large piston, while the large piston lifts 20 times the force. The work done on both sides is the same.

Unit 6: Gravitation

1. The motion of a falling ball towards Earth is due to the _____.

(a) Weightlessness

✓ (b) Gravitational force

(c) Acceleration due to gravity

(d) Both 'a' and 'b'

2. Newton's law of gravitation holds between every two objects on the _____.

(a) on Earth

(b) on Jupiter

(c) on Moon

✓ (d) on Universe

3. The numerical value of G is _____.

✓ (a) G = 6.673 x 10^-11 Nm2 kg^-2

(b) G = 6.673 x 10^11 Nm2 kg^-2

(c) G = 6.763 x 10^-10 Nm2 kg^-2

(d) G = 6.763 x 10^11 Nm2 kg^-2

4. The gravitational field of Earth is directed _____.

✓ (a) towards the Earth

(b) towards the Sun

(c) towards the Moon

(d) away from Earth

5. _____ was the first scientist who gave the concept of gravitation.

(a) Einstein

✓ (b) Newton

(c) Faraday

(d) Maxwell

6. According to Newton's law of universal gravitation, force is proportional to _____. (a) m1 m2 (b) 1 / r2 (c) r2 (d) Both (a) and (b)

7. Gravitational force is always _____.

(a) Repulsive

✓ (b) Attractive

(c) Both a and b

(d) None of these

8. The numerical value of _____ remains constant everywhere.

(a) g

✓ (b) G

(c) F

(d) W

9. Gravitation force is _____ of the medium between the objects.

(a) Dependent

✓ (b) Independent

(c) Both 'a' and 'b'

(d) None of these

10. Near Earth's surface (g) = _____.

✓ (a) 10 ms^-2

(b) 1.6 ms^-1

(c) Both

✓ (a) and

(b)

(d) None of these

11. Newton's law of gravitation is consistent with Newton's _____ law of motion.

(a) 1st

(b) 2nd

(c) 3rd

✓ (d) All of them

12. A spring balance is used to measure _____.

(a) Mass

✓ (b) Weight

(c) Elasticity

(d) Density

13. Your weight as measured on Earth will be _____ on the Moon.

(a) Increased

✓ (b) Decreased

(c) Remains same

(d) None of these

14. Mass of Earth is _____.

(a) 6.0 x 10^23 kg

✓ (b) 6.0 x 10^24 kg

(c) 6.0 x 10^25 kg

(d) 6.0 x 10^26 kg

15. _____ is a natural satellite.

(a) Earth

(b) Jupiter

✓ (c) Moon

(d) Mars

16. A communication satellite completes its one revolution around the Earth in _____ hours.

(a) 6

(b) 12

(c) 18

(d) 24

17. The velocity of a satellite is _____ of its mass.

(a) Independent

(b) Dependent

(c) Equal

(d) Double

18. _____ are used to put satellites into orbits.

(a) Helicopter

(b) Aeroplane

(c) Rocket

(d) None of these

19. The critical velocity (vc) = _____.

(a) gR

(b) g / R

Question: Section (B) Structured Questions. Newton's Law of Gravitation.

1. (a) Why do we not feel the gravitational force of attraction from the objects around us? (b) Define gravitational field with an example.

Answer: (a) Because the masses of everyday objects are very small, the gravitational force between them (F = G m1 m2 / r2) is extremely tiny compared to the Earth's pull, so we cannot feel it. (b) A gravitational field is the region around a mass in which another mass feels a gravitational force. Example: the region around the Earth — a ball released near the Earth is pulled towards it.

Question: Newton's Law of Gravitation. 2. (a) Write down any three characteristics of gravitational force. (b) Define gravitational field strength.

Answer: (a) (i) It is always attractive, never repulsive. (ii) It acts between any two masses anywhere in the universe. (iii) It is independent of the medium between the masses and follows the inverse-square law (it

weakens as 1 / r2). (b) Gravitational field strength at a point is the gravitational force per unit mass at that point: g = F / m. Its unit is N/kg (the same as ms^-2).

Question: Newton's Law of Gravitation. 3. (a) State and explain Newton's law of gravitation. (b) Define 'field force'.

Answer: (a) Newton's law of gravitation: every two masses in the universe attract each other with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them: F = G m1 m2 / r2, where G is the universal gravitational constant. (b) A field force is a force that acts at a distance through a field, without direct contact — for example, the gravitational force.

Question: Newton's Law of Gravitation. 4. Determine the gravitational force of attraction between Urwa and Ayesha standing at a distance of 50 m apart. The mass of Urwa is 60 kg and that of Ayesha is 70 kg.

Answer: F = G m1 m2 / r2 = 6.673 x 10^-11 x 60 x 70 / 50^2 = 6.673 x 10^-11 x 4200 / 2500 = 6.673 x 10^-11 x 1.68 = 1.12 x 10^-10 N.

Question: Weight. 5. (a) Why does the weight of an object not remain the same everywhere on Earth? (b) Why is the unit of weight Newton? Explain.

Answer: (a) Weight is W = mg, and g is not the same everywhere on Earth: it is slightly greater at the poles than at the equator and it decreases with height above the surface, so weight changes from place to place. (b) Weight is a force (the Earth's gravitational pull on the mass), and the SI unit of force is the newton.

Question: Weight. 6. (a) Define weight and write down its equation. (b) The weight of Rani is 450 N at the surface of Earth. Find her mass.

Answer: (a) Weight is the gravitational force with which the Earth attracts a body: W = mg. (b) m = W / g = 450 / 10 = 45 kg.

Question: Weight. 7. The weight of Naveera is 700 N on the Earth's

surface. What will be Naveera's weight at the surface of the Moon?

Answer: The Moon's g is about one-sixth of the Earth's g, so W_moon = 700 / 6 = 116.7 N (about 117 N).

Question: Weight. 8. (a) Your weight decreases as you go up to high altitudes, without dieting. Explain. (b) If you step on a scale and it gives a reading of 55 kg, is that a measure of your weight? If not, then which physical quantity does it show?

Answer: (a) Weight is W = mg and g = GM / r2, where r is the distance from Earth's centre. At high altitude r is larger, so g is smaller and your weight decreases, although your mass stays the same. (b) No. The scale shows mass (55 kg). A bathroom scale measures the force (weight) pressing on it but is calibrated to display mass in kilograms; your weight is W = mg = 55 x 10 = 550 N.

Question: Mass of Earth. 9. Calculate the mass of the Earth by using Newton's law of gravitation.

Answer: From g = G M / R2, M = g R2 / G = 10 x (6.4 x 10^6)^2 / (6.673 x 10^-11) = 10 x 4.096 x 10^13 / (6.673 x 10^-11) = 6.14 x 10^24 kg (about 6.0 x 10^24 kg).

Question: Mass of Earth. 10. If M_E is the mass of Earth, R_E the radius of Earth, and G the universal gravitational constant, then find the acceleration due to gravity g: (i) on the surface of Earth, (ii) at the centre of Earth.

Answer: (i) On the surface: g = G M_E / R_E^2. (ii) At the centre: g = 0. At the centre the mass pulls equally in all directions, so the net gravitational force — and hence g — is zero.

Question: Mass of Earth. 11. A planet has a mass four times that of Earth and a radius two times that of Earth. If the value of g on the surface of Earth is 10 ms^-2, calculate the acceleration due to gravity on the planet.

Answer: g_planet = G (4M) / (2R)^2 = 4GM / 4R^2 = GM / R^2 = g_earth = 10 ms^-2.

Question: Mass of Earth. 12. Evaluate the acceleration due to gravity in terms of the mass of Earth M_E, the radius of Earth R_E and the universal gravitational constant G: (i) at a distance twice the Earth's radius, (ii) at a distance one half the Earth's radius.

Answer: (i) At distance r = 2 R_E from the centre: g = G M_E / (2 R_E)^2 = G M_E / (4 R_E^2) = g_surface / 4. (ii) At distance r = R_E / 2 from the centre: g = G M_E / (R_E / 2)^2 = 4 G M_E / R_E^2 = 4 g_surface. [Note: the formula g = GM / r2 applies outside the Earth; for a point inside the Earth at half the radius, only the mass within that radius pulls, giving g / 2.]

Question: Artificial Satellite. 13. (a) Calculate the speed of a satellite which orbits the Earth at an altitude of 400 kilometers above Earth's surface. (b) Write the name of any one natural satellite.

Answer: (a) v = sqrt(GM / (R + h)) = sqrt(g R2 / (R + h)) = sqrt(10 x (6.4 x 10^6)^2 / (6.8 x 10^6)) = sqrt(6.02 x 10^7) = 7760 ms^-1 (about 7.8 km/s). (b) The Moon.

Question: Artificial Satellite. 14. (a) Write down the names of four different types of orbit. (b) Define the terms (i) Critical Velocity, (ii) Communication Satellite.

Answer: (a) Low Earth Orbit (LEO), Medium Earth Orbit (MEO), Geostationary / Geosynchronous Orbit (GEO), Polar Orbit. (b) (i) Critical velocity is the minimum horizontal velocity needed to place a satellite in a stable circular orbit around the Earth: vc = sqrt(gR), about 7.9 km/s for Earth. (ii) A communication satellite is a satellite placed in orbit (usually geostationary) to relay radio, television, telephone and internet signals between distant places on Earth.

Question: Artificial Satellite. 15. Derive the expression for the motion of a satellite: v = sqrt(GM / (R + h)).

Answer: For a satellite of mass m in a circular orbit of radius (R + h), the centripetal force is supplied by gravity: mv2 / (R + h) = G M m / (R + h)^2. Cancelling m and one factor (R + h): v2 = GM / (R + h), so v = sqrt(GM / (R + h)).

Question: Artificial Satellite. 16. (a) Differentiate between a natural satellite and an artificial satellite. (b) Name the parameters on the basis of which orbits are characterized.

Answer: (a) A natural satellite is formed by nature and orbits a planet by itself (e.g. the Moon around the Earth). An artificial satellite is man-made, launched by a rocket, and placed in orbit for a specific purpose such as communication, weather watch or navigation. (b) Orbits are characterized by their altitude (height above the Earth), orbital period, inclination (tilt of the orbital plane), and shape (circular or elliptical, i.e. eccentricity).

Unit 7: Properties of Matter

1. An object with particles close together and vibrating describes a _____.

(a) Gas

(b) Liquid

✓ (c) Solid

(d) All three

2. A burning candle is an example of _____ state of matter.

(a) Gas

(b) Liquid

(c) Solid

✓ (d) All three

3. During which process does a gas become a liquid _____?

(a) Melting

(b) Freezing

✓ (c) Condensing

(d) Boiling

4. A solid can _____.

✓ (a) have a fixed shape

(b) be easily compressed

(c) take the shape of its container

5. According to kinetic molecular theory, the pressure exerted by a gas is caused by the _____.

✓ (a) bombardment of the gas molecules on the walls of the container

(b) collision between gas molecules

(c) large distances between gas molecules

(d) random motion of the gas molecules

6. If a gas is heated in a sealed cylinder, then _____ increases.

(a) pressure inside the container

(b) average kinetic energy of the particles

(c) temperature of the gas

✓ (d) All of them

7. A gas in a container of fixed volume is heated. What happens to the molecules of the gas?

(a) They collide less frequently.

(b) They expand.

✓ (c) They move faster.

(d) They move further apart.

8. In a liquid, some energetic molecules break free from the surface even when the liquid is too cold for bubbles to form. What is the name of this process?

(a) boiling

(b) condensation

(c) convection

✓ (d) evaporation

9. What happens to the molecules of a gas when the gas changes into a liquid?

✓ (a) They move closer and lose energy.

(b) They move closer and gain energy.

(c) They move apart and lose energy.

(d) They move apart and gain energy.

10. A substance has a melting point of -17 degrees C and a boiling point of 117 degrees C. In which state does the substance exist at -10 degrees C and at 110 degrees C?

(a) Solid at -10, liquid at 110

(b) Solid at -10, gas at 110

(c) Liquid at -10, liquid at 110

(d) Liquid at -10, gas at 110

Question: Section (B) Structured Questions. Kinetic molecular model of matter. 1. (a) "The particles are free to move within the material, it has a fixed volume but takes up the shape of its container" — which state of matter is being described here? (b) Write a similar description of the particles that make up a solid. (c) Write down any two properties of a solid.

Answer: (a) Liquid. (b) In a solid the particles are closely packed in fixed positions; they cannot move freely and can only vibrate about their mean positions. (c) Any two: fixed shape; fixed volume; incompressible; high density; cannot be poured.

Question: Kinetic molecular model of matter. 2. (a) Why is the kinetic model of matter called kinetic? (b) In which state of matter are the molecules widely separated? (c) In which state of matter are the molecules most closely packed? (d) In which state of matter can molecules move freely at high speed?

Answer: (a) It is called kinetic because it explains matter by the motion (kinetic energy) of its particles: the particles of all matter are always moving. (b) Gas. (c) Solid. (d) Gas.

Question: Kinetic molecular model of matter. 3. (a) By using kinetic molecular theory, explain why we can walk through air, swim through water, but cannot walk through a solid wall. (b) In which state of matter do the molecules have minimum kinetic energy? (c) Which state of matter is highly incompressible?

Answer: (a) In air the molecules are far apart with very weak forces between them, so we push through easily. In water the molecules are closer but still free to move past one another, so we can swim through. In

a solid wall the molecules are tightly packed in fixed positions with strong forces, so we cannot pass through. (b) Solid. (c) Solid (liquids are also highly incompressible).

Question: Forces and Kinetic Theory. 4. A sample of a gas in a sealed test tube is cooled. Describe what happens to: (a) the size of the molecules, (b) the speed at which molecules move, (c) the number of the molecules, (d) the pressure inside the tube, (e) the state of the gas.

Answer: (a) The size of the molecules does not change. (b) The molecules move more slowly (their kinetic energy falls). (c) The number of molecules stays the same (the tube is sealed). (d) The pressure inside the tube falls (slower, less energetic collisions with the walls). (e) If cooled enough, the gas condenses into a liquid and may then freeze into a solid.

Question: Forces and Kinetic Theory. 5. An inflated car tyre is considered to have a constant volume, regardless of any changes in temperature or pressure. Use the kinetic molecular theory of gases to answer: (a) How does the air in the tyre exert a pressure on the walls of the tyre? (b) Why is the pressure the same at all points on the inside wall of the tyre? (c) More air is pumped into the tyre while the temperature is kept constant until there are twice as many molecules as before. Explain why you would expect the pressure to be doubled.

Answer: (a) The air molecules move rapidly in all directions and collide with (bombard) the inner walls; these collisions exert the pressure. (b) The molecules are spread evenly and move randomly in all directions, so collisions happen equally often at every point of the wall, giving the same pressure everywhere. (c) With twice as many molecules at the same temperature, there are twice as many collisions per second with the walls, so the pressure doubles (at fixed volume and temperature, pressure is proportional to the number of molecules).

Question: Forces and Kinetic Theory. 6. Describe the following: (a) What happens to the motion of the molecules of a gas when it cools down? (b) What happens to the motion of a liquid when it cools down?

Answer: (a) Gas: the molecules slow down as their kinetic energy falls; if cooled enough they come closer together and the gas condenses into a liquid. (b) Liquid: the molecules slow down and move less freely; on further cooling the liquid freezes into a solid, where molecules only vibrate in fixed positions.

Question: Gases and Kinetic theory. 7. The pressure on 9 cm3 of oxygen gas is doubled at a fixed temperature. What will its volume become?

Answer: By Boyle's law, P1 V1 = P2 V2. Doubling the pressure halves the volume: V2 = 9 / 2 = 4.5 cm3.

Question: Gases and Kinetic theory. 8. A container holds 30 m3 of air at a pressure of 150000 Pa. If the volume is changed to 10 m3 by decreasing the load on the piston, what will the pressure of the gas become? Assume that its temperature remains constant.

Answer: By Boyle's law, P2 = P1 V1 / V2 = 150000 x 30 / 10 = 450000 Pa. [Note: the book says the volume is changed "by decreasing load on the piston", but a smaller volume needs a greater load; the numbers are used as printed.]

Question: Gases and Kinetic theory. 9. Air at atmospheric pressure of 760 mm of Hg is trapped inside a container fitted with a moveable piston. When the piston is pulled out slowly so that the volume is increased from 100 dm3 to 150 dm3, the temperature remaining constant, what will the pressure of the air become?

Answer: By Boyle's law, P2 = P1 V1 / V2 = 760 x 100 / 150 = 506.7 mm of Hg (about 507 mm Hg).

Unit 8: Energy Sources and Transfer of Energy

1. If a force of 6 N displaces an object 2 m in the direction of the force, then the work done will be _____.

(a) 0 Joule

✓ (b) 12 Joule

(c) 3 Joule

(d) 12 N

2. If a body of mass 1 kg is moving with a velocity of 1 m/sec, then the K.E of the body will be _____.

(a) 10 Joules

✓ (b) 0.5 Joules

(c) 2 Joules

(d) 1 Joule

3. If a machine performs 20 J of work in 10 sec, then its power is _____.

(a) 200 watt

(b) 20 watt

✓ (c) 2 watt

(d) 0.2 watt

4. A body of mass 1 kg is lifted through a height of 1 m. The energy possessed by the body will be _____ (consider g = 10 ms^-2).

(a) 1 J

✓ (b) 10 Joule

(c) 100 Joule

(d) 1000 Joule

5. The energy released during a fission or fusion reaction is called _____.

(a) Solar energy

(b) Geothermal energy

(c) Tidal energy

✓ (d) Nuclear energy

6. Which is the renewable source of energy?

✓ (a) Solar and wind

(b) Coal

(c) Natural gas

(d) Petroleum

7. The ratio of output to input is called _____.

(a) Energy

(b) Work

(c) Power

✓ (d) Efficiency

8. Work done per unit time is called _____.

(a) Efficiency

(b) Energy

✓ (c) Power

(d) Force

9. Coal, gas and oil are all examples of _____.

(a) Tidal energy

(b) Nuclear energy

✓ (c) Fossil fuel energy

(d) Biomass energy

10. _____ is not a renewable source of energy.

(a) Solar energy

✓ (b) Coal

(c) Wind energy

(d) Geothermal energy

Question: Section (B) Structured Questions. Work. 1. (a) Define work. (b) Derive the equation: Work = F d cos theta.

Answer: (a) Work is done when a force moves a body through a distance in the direction of the force: W = F x d. (b) If the force F acts at an angle theta to the displacement d, only the component of force along the displacement, F cos theta, does work. So W = (F cos theta) x d = F d cos theta.

Question: Work. 2. How much work is needed to move horizontally a body 20 m by a force of 30 N, if the angle between the force and the horizontal surface is 60 degrees?

Answer: W = F d cos theta = 30 x 20 x cos 60 = 30 x 20 x 0.5 = 300 J.

Question: Work. 3. How much work is done if a crate is moved a distance of 50 m when a force of 30 N is applied along the surface?

Answer: W = F x d = 30 x 50 = 1500 J.

Question: Work. 4. What is the work done by Usman? If a bar of weight 100 N is brought by him from (A) to (B), then brought back to (A).

Answer: The net displacement is zero (the bar ends where it started), so the net work done is zero: W = F x 0 = 0 J. (Work done lifting/carrying in one direction is cancelled on the return.)

Question: Energy Forms. 5. (a) Define kinetic energy. (b) Derive the equation (K.E = 1/2 m v2).

Answer: (a) Kinetic energy is the energy a body has because of its motion. (b) Work done to speed up a mass m from rest to velocity v: W = F x d = m a x d. From v2 = 2 a d, d = v2 / 2a. So W = m a x (v2 / 2a) = 1/2 m v2. This work is stored as kinetic energy: K.E = 1/2 m v2.

Question: Energy Forms. 6. What will be the kinetic energy of a boy of mass 50 kg driving a bike with a velocity of 2 ms^-1?

Answer: K.E = 1/2 m v2 = 1/2 x 50 x 2^2 = 1/2 x 50 x 4 = 100 J.

Question: Energy Forms. 7. (a) Define potential energy. (b) Derive the equation (P.E = mgh).

Answer: (a) Potential energy is the energy stored in a body because of its position or state. (b) Work done to lift a mass m through height h against gravity: W = F x d = (mg) x h = mgh. This work is stored as potential energy: P.E = mgh.

Question: Energy Forms. 8. (a) If an LED screen of mass 10 kg is lifted up and kept on a cupboard of height 2 m, calculate the potential energy stored in the LED screen. (b) Calculate the potential energy of 3 kg water raised to the tank at the roof of a home 4 m high. (assume g = 10 ms^-2)

Answer: (a) P.E = mgh = 10 x 10 x 2 = 200 J. (b) P.E = mgh = 3 x 10 x 4 = 120 J.

Question: Conversion of Energy. 9. (a) Why is fossil fuel energy called a non-renewable source? (b) Define solar energy and its importance in Pakistan.

Answer: (a) Fossil fuels (coal, oil, natural gas) took millions of years to form from dead plants and animals, and we are using them far faster than nature can replace them. Once used up, they cannot be replaced within a human lifetime. (b) Solar energy is the energy we get from the Sun's light and heat. Importance in Pakistan: Pakistan gets strong sunshine almost all year; solar power can supply electricity, reduce load-shedding, serve remote areas without grid lines, and cut fuel imports and pollution.

Question: Conversion of Energy. 10. Write notes on tidal energy and geothermal energy.

Answer: Tidal energy: energy from the rise and fall of sea tides, caused by the Moon's (and Sun's) gravitational pull. Tidal barrages and underwater turbines turn this moving water into electricity. Geothermal energy: heat from inside the Earth (hot rocks and magma below the surface). It is used

to make electricity (steam turbines) and for direct heating.

Question: Conversion of Energy. 11. (a) What is wind energy? (b) Write any three applications of wind energy.

Answer: (a) Wind energy (the book prints "wing energy") is the energy of moving air. (b) Any three: generating electricity with wind turbines; pumping water from wells; grinding grain with windmills; driving sailing ships.

Question: Conversion of Energy. 12. (a) Write the name of any one radioactive element which is used as a source of nuclear energy. (b) Write the name of any one device that can convert solar energy into heat energy. (c) Write the names of any two devices that can convert solar energy into electrical energy.

Answer: (a) Uranium (Uranium-235). (b) Solar water heater (or solar cooker). (c) Solar cell and solar panel (photovoltaic panel).

Question: Renewable and Non-renewable Energy Sources. 13. Write a note on renewable energy sources.

Answer: Renewable energy sources are those that nature keeps replacing, so they never run out on human timescales: solar, wind, hydroelectric (flowing water), tidal, geothermal and biomass energy. They are clean and sustainable, and using them reduces pollution and dependence on fossil fuels.

Question: Renewable and Non-renewable Energy Sources. 14. Write a note on non-renewable energy sources.

Answer: Non-renewable energy sources exist in limited amounts and are used up much faster than nature can replace them: coal, oil (petroleum), natural gas and nuclear fuel (uranium). Once finished, they cannot be replaced within a human lifetime. Burning fossil fuels also causes air pollution and climate change.

Question: Renewable and Non-renewable Energy Sources. 15. What is the difference between renewable and non-renewable energy sources?

Answer: Renewable sources are replaced by nature continuously and never run out (solar, wind, hydro, tidal, geothermal, biomass); they are generally clean. Non-renewable sources are finite: once used up they are gone for human purposes (coal, oil, natural gas, uranium), and burning fossil fuels pollutes the air.

Question: Renewable and Non-renewable Energy Sources. 16. Make a table of renewable and non-renewable energy sources from the following: Uranium, Solar, Coal, Wind, Natural gas, Tidal, Biomass, Hydroelectricity.

Answer: Renewable: Solar, Wind, Tidal, Biomass, Hydroelectricity.
Non-renewable: Uranium, Coal, Natural gas.

Question: Efficiency. 17. Calculate the efficiency of a machine which consumes 200 J of energy and performs 50 J of work.

Answer: Efficiency = (output / input) x 100 = (50 / 200) x 100 = 25%.

Question: Efficiency. 18. Write a note on efficiency.

Answer: Efficiency is the ratio of the useful output of a machine to its total input, given as a percentage: efficiency = (output / input) x 100. It is always less than 100% because some input energy is always lost as heat, sound or friction.

Question: Efficiency. 19. If the efficiency of a machine is 70% and its output is 100 J, then calculate its input.

Answer: Input = output / efficiency = 100 / 0.70 = 142.9 J (about 143 J).

Question: Efficiency. 20. Which machine is more efficient: machine "A" which has an output of 200 J after consuming 400 J of energy, or machine "B" which has an output of 300 J after consuming 450 J of energy?

Answer: Machine A: 200 / 400 x 100 = 50%. Machine B: 300 / 450 x 100 = 66.7%. Machine B is more efficient.

Question: Power. 21. (a) Define power. (b) The energy of 600 J is dissipated by a bulb in 50 minutes. Find the power of the bulb.

Answer: (a) Power is the rate of doing work (or of transferring energy): P = W / t. (b) Time = 50 x 60 = 3000 s. P = 600 / 3000 = 0.2 W.

Question: Power. 22. (a) Convert 20 watt into horse power. (b) Calculate the power of a machine if it does 40 Joules of work in 10 sec.

Answer: (a) 1 hp = 746 W, so 20 W = 20 / 746 = 0.0268 hp. (b) P = W / t = 40 / 10 = 4 W.

Question: Power. 23. (a) Define Watt. (b) A student of weight 400 N takes 5 sec to climb up an obstacle of height 2 m. Calculate the power consumed.

Answer: (a) Watt is the SI unit of power: 1 watt = 1 joule per second (1 W = 1 J/s). (b) Work = F x h = 400 x 2 = 800 J. Power = 800 / 5 = 160 W.

Question: Power. 24. (a) Write down the names of any two larger units of power. (b) If a machine consumes 250 J of energy per hour, then what will be its power?

Answer: (a) Kilowatt (kW) and megawatt (MW); horsepower (hp) is also a larger unit. (b) P = E / t = 250 / 3600 = 0.0694 W (about 0.07 W).

Unit 9: Thermal Properties of Matter

1. The internal energy of a system is the sum of its _____.

(a) Heat

(b) Kinetic energy

✓ (c) Energy

(d) None of these

2. The amount of heat required to raise the temperature of an object through 1 degree Celsius depends on _____.

(a) The mass of the object

(b) The temperature of the object

(c) The shape of the object

✓ (d) Specific heat of material

3. The heat energy needed to change a liquid into gas without changing its temperature is called _____.

✓ (a) Heat of Vaporization

(b) Heat of Fusion

(c) Specific Heat

(d) Latent Heat

4. The amount of heat energy needed to raise the temperature of 1 kg of a substance by 1 degree Celsius is called its _____.

(a) Thermal capacity

✓ (b) Specific heat capacity

(c) Latent heat of fusion

(d) Latent heat of vaporization

5. The temperature at which a liquid starts to change into gas is called _____.

(a) Melting point

(b) Freezing point

✓ (c) Boiling point

(d) Condensation point

6. The melting point of ice is _____.

✓ (a) 0 degrees C

(b) 100 degrees C

(c) -10 degrees C

(d) -273 degrees C

7. Heat absorbed by a substance in changing state from solid to liquid without change in temperature is called _____.

✓ (a) Latent heat of fusion

(b) Latent heat of vaporization

(c) Specific heat capacity

(d) Heat capacity

8. Heat absorbed by a substance in changing state from liquid to gas without change in temperature is called _____.

(a) Latent heat of fusion

✓ (b) latent heat of vaporization

(c) Specific heat capacity

(d) Heat capacity

9. Vaporization takes place at _____.

(a) boiling point

(b) melting point

(c) freezing point

✓ (d) all temperatures

10. The rate of evaporation is increased by _____.

(a) increasing humidity

(b) decreasing temperature

(c) increasing boiling point

✓ (d) decreasing atmospheric pressure

11. The factor(s) which increase the rate of evaporation are _____.

(a) increase in surface area

(b) increase in temperature

(c) increase in wind speed

✓ (d) all of these

Question: Section (B) Structured Questions. Thermal energy, temperature and heat. 1. (a) Explain the difference between thermal energy and heat. (b) Define heat capacity and write down its unit.

Answer: (a) Thermal energy is the total kinetic energy of all the particles of a body. Heat is thermal energy in transfer: it flows from a hotter body to a colder one because of a temperature difference. (b) Heat capacity is the heat needed to raise the temperature of a whole body by 1 degree C: C = Q / delta T. Its unit is J per degree C (J / degrees C) or J/K.

Question: Thermal energy, temperature and heat. 2. (a) Define specific heat capacity and write down its unit. (b) What is the value of the specific heat of water?

Answer: (a) Specific heat capacity is the heat needed to raise 1 kg of a substance by 1 degree C: c = Q / (m delta T). Its unit is J kg^-1 degrees C^-1 (J per kg per degree C). (b) The specific heat of water is 4200 J kg^-1 degrees C^-1.

Question: Thermal energy, temperature and heat. 3. One litre of water is heated by 4 degrees C to 100 degrees C. How much heat energy is supplied?

Answer: One litre of water has a mass of 1 kg. Q = m c delta T = 1 x 4200 x 4 = 16800 J.

Question: Thermal energy, temperature and heat. 4. If heat energy supplied to 2 kg ice is increased by 15 degrees C, how much heat energy is absorbed by ice? (Assume specific heat capacity of ice is 1000 J kg^-1 C^-1)

Answer: Q = m c delta T = 2 x 1000 x 15 = 30000 J.

Question: Latent heat. 5. (a) Define heat of fusion. (b) Define heat of vaporization. (c) Write down the formula for latent heat.

Answer: (a) Heat of fusion is the heat needed to change 1 kg of a solid into liquid at its melting point without changing its temperature. (b) Heat of vaporization is the heat needed to change 1 kg of a liquid into gas at its boiling point without changing its temperature. (c) Q = m L, where L is the latent heat.

Question: Latent heat. 6. How much heat is needed to convert 5 kg of water into steam at its boiling point?

Answer: Using the book's latent heat of vaporization of water, L = 2260 kJ kg^-1: Q = m L = 5 x 2260 = 11300 kJ.

Question: Latent heat. 7. (a) Define latent heat. (b) How much heat is needed to convert (i) 1 kg of water to steam at its boiling point, (ii) 5 kg of water to steam at its boiling point?

Answer: (a) Latent heat is the heat absorbed or released when a substance changes its state without any change in temperature. (b) With L = 2260 kJ kg^-1: (i) Q = 1 x 2260 = 2260 kJ. (ii) Q = 5 x 2260 = 11300 kJ.

Question: Evaporation and boiling. 8. (a) Define evaporation. (b) Define boiling. (c) Differentiate between evaporation and boiling.

Answer: (a) Evaporation is the escape of molecules from the surface of a liquid at any temperature. (b) Boiling is the rapid change of a liquid into gas throughout the liquid, at a fixed temperature (the boiling point). (c) Evaporation happens at all temperatures and only at the surface, and is

slow; boiling happens only at the boiling point, throughout the liquid, and is fast.

Question: Thermal expansion. 9. (a) Define thermal expansion. (b) Explain the expansion of a solid on heating. (c) Write a note on the anomalous expansion of water.

Answer: (a) Thermal expansion is the increase in the size (length, area or volume) of a body when it is heated. (b) On heating, the particles of a solid vibrate more vigorously about their positions and push each other further apart, so the solid expands. (c) Anomalous expansion of water: water contracts when heated from 0 degrees C to 4 degrees C and expands above 4 degrees C. Its volume is smallest (density greatest) at 4 degrees C. Because of this, ice floats on water and lakes freeze from the top down, letting aquatic life survive below.

Question: Thermal energy, temperature and heat. 10. Find the heat absorbed by a 5 kg metal if its temperature is increased from 20 degrees C to 80 degrees C. The specific heat of the metal is 0.39 kJ kg^-1 C^-1.

Answer: Q = m c delta T = 5 x 0.39 x (80 - 20) = 5 x 0.39 x 60 = 117 kJ.

Question: Thermal energy, temperature and heat. 11. How much energy is transferred when 1.5 kg of metal is heated from 10 degrees C to 60 degrees C, if the heat absorbed by the metal is 5000 J? Find the specific heat capacity of the metal.

Answer: The energy transferred is the heat absorbed = 5000 J. c = Q / (m delta T) = 5000 / (1.5 x 50) = 5000 / 75 = 66.7 J kg^-1 C^-1.

Question: Thermal energy, temperature and heat. 12. Find the specific heat capacity of a 5 kg metal whose temperature is raised by 25 degrees C when 500 J of heat energy is supplied to it.

Answer: c = Q / (m delta T) = 500 / (5 x 25) = 500 / 125 = 4 J kg^-1 C^-1. [Note: this is far below the specific heat of any real metal (hundreds of J

kg^-1 C^-1); the book's numbers give 4, which suggests a misprint in the question.]

Question: Thermal energy, temperature and heat. 13. A 200 g piece of metal is heated through a temperature rise of 5 degrees C. How much heat is needed? (Specific heat capacity of water is 4200 J kg^-1 C^-1.)

Answer: Taking the specific heat as given (4200 J kg^-1 C^-1): Q = m c delta T = 0.2 x 4200 x 5 = 4200 J.

Question: Thermal energy, temperature and heat. 14. A 0.5 kg substance is heated from 20 degrees C to 70 degrees C. If 12500 J of heat energy is absorbed, find the specific heat capacity of the substance.

Answer: c = Q / (m delta T) = 12500 / (0.5 x 50) = 12500 / 25 = 500 J kg^-1 C^-1.

Question: Thermal energy, temperature and heat. 15. (a) Find the heat capacity of a 2.5 kg block of metal if the temperature rises by 20 degrees C when 900 J of energy is supplied. (b) Find the specific heat capacity of the metal.

Answer: (a) Heat capacity C = Q / delta T = 900 / 20 = 45 J per degree C. (b) Specific heat capacity c = C / m = 45 / 2.5 = 18 J kg^-1 C^-1. [Note: this is far below the specific heat of any real metal; the book's numbers give 18.]